

Rakib Mahmud

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EDUCATION

• Khulna University of Engineering & Technology, (KUET)

November 2018 - March 2024

B.Sc. in Computer Science and Engineering

Khulna, Bangladesh

◦ CGPA: 3.71/4.00

RESEARCH INTERESTS

- Artificial Intelligence
- Computer Vision
- Federated learning
- Deep Learning
- Natural Language Processing
- Machine Learning and Unlearning

RESEARCH EXPERIENCE

◦ Multi-Class Chest Disease Classification from X-Rays Using PCA-Enhanced Vision Transformers 2025 - 2026

Advisor: Dr. Ch. Md. Rakin Haider, Dept. of Computer Science & Engineering, BUET

- * Developed a deep learning framework for multi-class chest disease classification (COVID-19, Pneumonia, Tuberculosis, Normal) from X-ray images, using PCA to reduce feature redundancy and computational cost while preserving discriminative information.
- * Implemented a hybrid ensemble of Vision Transformer (locality-aware self-attention) and EfficientNet to capture global and local radiographic patterns, with Grad-CAM and LIME for clinical interpretability.

◦ A Hybrid Deep Learning Approach with Explainable AI for Bangladeshi Birds Classification 2023 - 2024

Thesis Supervisor: Dr. Sk Md Masudul Ahsan, Dept. of Computer Science & Engineering, KUET

- * Built a pipeline using YOLOv8 for bird localization and transfer learning for classification on a self-curated dataset.
- * Designed a hybrid CNN with Lab color space histogram features for limited computational resources, and integrated Grad-CAM to enhance interpretability for biodiversity applications.

◦ Analysis of Biogeochemical Indicators in the Southern Ocean 2022 - 2023

Advisor: Sunanda Das, Dept. of Computer Science & Engineering, KUET

- * Forecasted key Southern Ocean indicators such as pH (25 °C) and nitrate concentration using SVR with correlation-based feature selection for climate monitoring applications.
- * Proposed novel custom kernels (Hyperbolic Sine and Gaussian Matrix Multiplier) for SVR with superior predictive performance over standard kernels.

TEACHING EXPERIENCES

◦ Bangladesh University of Business and Technology (BUBT)

July 2025 - Present

Lecturer, Dept. of Computer Science & Engineering

Dhaka, Bangladesh

- * Taught undergraduate courses and labs on Data Structures, Software Engineering, and Pattern Recognition.
- * Advised and mentored undergraduate cohorts; served as Course Coordinator for Pattern Recognition.

◦ Dhaka International University (DIU)

July 2024 - July 2025

Lecturer, Dept. of Computer Science & Engineering

Dhaka, Bangladesh

- * Taught courses and labs on Discrete Mathematics, Structured and Object-Oriented Programming, Compiler Design, Computer Architecture, Markup and Scripting Languages, and Software Engineering.
- * Served as Batch Counselor (meeting documentation, academic tour coordination, research publication assistance) and Co-Moderator of the DIU Film and Photography Club.

PUBLICATIONS

- [1] Ankan Saha, **Rakib Mahmud**, and Rifat Shahriyar. **"Deconstructing the Garbage-First Collector: A Python Simulation Study with an Adaptive Hot-Card Threshold."** 4th International Conference on Computing Advancements (ICCA 2026). ACM, 2026. (Accepted)
- [2] **Rakib Mahmud**, and Sk Md Masudul Ahsan. **"A Hybrid Deep Learning Approach with Explainable AI for Bangladeshi Birds Classification."** 2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM). IEEE, 2025.
- [3] Saad, Asif Mohammed, **Rakib Mahmud**, and Sunanda Das. **"Forecasting Crucial Biogeochemical Indicators of the Southern Ocean for Climate Monitoring Using Modified Kernel-Based Support Vector Regression."** International Conference on Big Data, IoT and Machine Learning. Singapore: Springer Nature Singapore, 2023.

PROJECTS

- **End-to-End Sentiment Analysis System Using Traditional ML and Transformer Models**

Python, PyTorch, OpenCV, Scikit-Learn, Numpy, Pandas, Matplotlib



- * Built a sentiment analysis pipeline on a large-scale Reddit dataset, combining TF-IDF bigram features and class-imbalance handling with BERT classification for improved recall, with MLflow experiment tracking.

- **KUETPS Website Using PHP**

PHP, MS SQL, Bootstrap, CSS, HTML



- * Created KUETPS Photographic Society website with user authentication, image interaction (upload, like, dislike, comment), secure database storage, and profile customization.

- **Automobile Company's Database**

SQL, Oracle



- * Designed and implemented a comprehensive database system for automobile company operations with normalized schema, complex queries, and transaction management.

- **Custom Programming Language**

Flex, Bison



- * Developed a sophisticated compiler integrating lexical and syntax analysis, efficient code optimization and generation, robust error handling, and cross-platform compatibility.

- **Dot and Boxes Game**

Python



- * Developed dynamic Dot and Boxes game for human-vs-AI gameplay with Alpha-Beta Pruning for AI decision-making optimization, enhancing player experience.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java
- **Data Analysis & Visualization:** OpenCV, Scikit-Learn, NumPy, Pandas, Matplotlib
- **Databases:** Oracle, Microsoft SQL Server, Firebase
- **Web Programming:** HTML, CSS, JavaScript, PHP
- **Software Tools:** MS WORD, EXCEL, PowerPoint
- **Developer Tools:** Git, GitHub, OpenGL, LaTeX

HONOURS AND AWARDS

- **Dean's Award**

Khulna University of Engineering & Technology

2019-20 and 2018-19



- * Awarded for Sophomore and Freshman Year

- **Dutch-Bangla Bank Scholarship**

Dutch-Bangla Bank Foundation

2019-2024



- * Awarded for Eight Consecutive Semesters

EXTRA CURRICULUM ACTIVITIES

- **President**

KUET Photographic Society (KUETPS)

2023 - 2024



- * Led the creative photography community of KUET that nurtures visual storytelling through workshops, photo walks, training sessions, exhibitions, and competitions for photographers of all levels.

- **Member Secretary & Founding Member**

CSE KUET Research Society (KRS)

2023 - 2024



- * Contributed to KUET's first research-focused club by organizing workshops and seminars, mentoring students on research and academic writing.

REFERENCES

Dr. Sk. Md. Masudul Ahsan

Professor

Department of Computer Science & Engineering

Khulna University of Engineering & Technology (KUET)

Email: smahsan@cse.kuet.ac.bd

Sunanda Das

Assistant Professor

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